Sahaj Bhadja

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Summary

Software Engineer with 2+ years building production **Python** APIs, **multi-tenant** data platforms, and ML-serving systems, from async job orchestration and real-time **WebSockets** to transformer policy pipelines for autonomous driving.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Analysis of Algorithms, Advanced NLP, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Software Engineering, Data Retrieval, Object-Oriented Programming

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Architected a **multi-tenant** platform on Azure SQL and Blob Storage with row-level isolation for 500+ users, cutting dashboard latency from 1.2s to 350ms through indexed query and access-pattern optimization.
- Restructured async **model job orchestration** with FastAPI background workers, explicit job-state transitions, and failure isolation, processing 1K+ LLM jobs at 3x throughput under concurrent load without cascading failures.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Served multimodal models behind a **switchable provider API** routing between Qwen-VL, GPT-4o Vision, and Claude Vision with timeout fallback, handling 10K+ daily requests for 150+ instructors with stable latency.
- Built secure **data ingestion** for PDF, DOCX, and ZIP uploads with concurrent batch processing of 1K+ files and per-call token and cost logging persisted to MySQL on AWS RDS.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Built a GPT-based Decision Transformer **policy model** for offline RL in autonomous driving, with PyTorch, Hydra, and W&B pipelines over 100K+ trajectory logs, improving long-horizon planning 18% in simulation.
- Designed a spatial-temporal encoder fusing HD-map geometry and agent trajectories via multi-head attention, then **evaluated** it in closed-loop simulation against three encoder baselines, improving planning stability 15%.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Optimized large-scale APIs with indexing, API-layer **caching**, pagination, and field filtering, improving response times 30% for low-bandwidth clients and reducing technical debt 20% with modular object-oriented designs.
- Redesigned ETL on AWS Glue and S3, boosting pipeline efficiency 68%, and automated nightly Jenkins **CI/CD** with 20+ unit and integration suites across Windows, Linux, and macOS builds.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built eight Python FastAPI microservices on PostgreSQL with **JWT** authentication and role-based access control, streaming real-time **WebSocket** updates to clients while synchronizing 2M+ records across distributed locations.
- Shipped Docker services to **Kubernetes** on GCP through Jenkins CI/CD with rolling zero-downtime updates and approval gates, reaching 95%+ build validation and monitoring production errors with Sentry.

Skills

Languages: Python, TypeScript, JavaScript, SQL, Java, C++
Backend & Platform: FastAPI, Flask, REST APIs, WebSockets, Microservices, Async Job Orchestration, Caching, Multi-Tenancy
ML Systems: PyTorch, Transformers, Model Serving, Inference APIs, Fine-Tuning, Model Evaluation, Hydra, Weights & Biases
Data: PostgreSQL, SQL Server, Supabase, Blob Storage, S3, ETL Pipelines
Infrastructure: GCP, AWS, Azure, Docker, Kubernetes, CI/CD, GitHub Actions
Reliability, Security & UI: Observability, Sentry, Row-Level Security, JWT, RBAC, React

Projects

Multi-Tenant Model Fine-Tuning and Inference API | *Python, FastAPI, PostgreSQL, Redis, Kubernetes, GCP, WebSockets*

- Designed a multi-tenant model API with FastAPI, per-tenant keys and Redis **rate limiting**, validated dataset uploads, Kubernetes fine-tuning jobs, versioned models in Postgres, and **WebSocket** streaming inference with backpressure.

CaptionCraft Live | *PyTorch, Transformers, BLIP, GIT, ViT-GPT-2, Flask*

- Fine-tuned and benchmarked BLIP, GIT, and ViT-GPT-2 on 10K images and 100K captions, raising accuracy from 81% to 96%, and served the best model through real-time **inference APIs**.

Big-Brain Live | *GCP, PostgreSQL, FAISS, LLaMA-2, REST APIs*

- Built a cloud ingestion and indexing pipeline on GCP over 50K+ documents with incremental updates, metadata persistence in PostgreSQL, and FAISS embedding retrieval, reducing **retrieval latency** 40%.